One-sheet: the Feynman path sum at closure order $\alpha = 2$

An Axiomatic Derivation of the Feynman Path Sum at the Light-Cone Kernel Width and Closure Order $\alpha = 2$ of the Causal Accumulation Law. S. Leizerman, LIOR Research Initiative, November 13, 2026. Script: `cal_feynman_alpha2.py` (numpy only, about 25 s). Units: $\hbar = c = 1$, $\tau_c = 1/\sqrt{2}$, so $\mu = 1$.

Inputs.

- The CAL at $\alpha = 2$: $T = 2J + K_2 J$. The transport chain is 1, so there is no coupling constant.
- A light-cone kernel: at lag τ the profile is the light front of radius $c\tau$.
- The balance reading: a free carrier balances its light-cone past ($T = 0$), and a source is the imbalance. The edge's update rule is not used here.
- The phase belongs to the source. The double accumulation gives the half turn, and no factor goes on the memory.
- For the readout, one sign of frequency, inherited from the source's past.

The kernel's lag scale τ_c is the only scale; only ratios such as $\mu\tau_c = 1/\sqrt{2}$ are physical.

Result	**Statement**	**Check**
Lorentz selects $\alpha = 2$	The light-cone symbol depends on $s^2 + c^2k^2$ alone only at $\alpha = 2$ (and 0). The defect is $\alpha(2-\alpha)\varphi^2/6$.	defect 0 at $\alpha = 0, 2$; 0.10 at 1.9, 2.1
Kernel = Green's function	Kirchhoff's shell in $3+1$; one order-1 accumulation per null leg in $1+1$ ($2 = 1+1$).	closed form matches direct integration to 6 digits
Klein–Gordon	The balanced carrier obeys $(\Box + \mu^2)J = 0$ with $\mu\tau_c = 1/\sqrt{2}$. The mass is the kernel's own rate.	roots $= \sqrt{k^2+\mu^2}$ to 10 digits
Conservative at 2	Both signs decay at $\frac{\pi}{4}(2-\alpha)\mu^2/\omega$ below 2 and run away at that rate above. Nothing near 2 picks a sign.	$\mathrm{Im}\,\omega$: -5.5×10^{-3}, 0, $+5.6\times10^{-3}$ at 1.99, 2, 2.01
Path sum	The Neumann series is a sum over piecewise-null histories with $-\mu^2$ per corner. It converges absolutely and resums to G_μ.	$1+1$ residual $3.35\times10^{-5} \to 8.38\times10^{-6}$
Checkerboard	The n-th term is the $(2n+1)$-corner amplitude over $2ic\mu$, since $(-\mu^2)^n = (i\mu)^{2n}$.	lattice error $\propto \varepsilon$
Corner tail	In $3+1$, histories without corners sit on the cone; the corners alone build the J_1 tail inside it.	$\leq 2.5\times10^{-6}$
Unitarity, readout	Each sign of frequency is unitary. On one sign, $\int JJ^\dagger$ is conserved, so the readout is Born-normalized in every frame.	spread 2.5×10^{-16} vs 1.6×10^{-1}
Feynman's kernel	The positive-frequency path sum, rest phase removed, gives K_F as $c \to \infty$ with error $O(1/c^2)$.	error$\cdot c^2 \to 3.667$ (pred. 3.672)

Where the i comes from. The $\alpha = 2$ dynamics is real. The i of K_F enters only with the sign of frequency inherited from the source. At the edge it is the rotation produced by negative feedback.

Not claimed / open.

- Interaction between carriers (the return leg).
- Spin (a Dirac form in $3+1$).
- Mass ratios between carriers.
- The relation to the edge's rest rotation; the equivalence condition is a forward pointer only.
- A frame-independent position readout (Newton–Wigner).
- Prior art: the $1+1$ checkerboard (Feynman–Hibbs, Jacobson–Schulman, GJKS) and causal-set chain sums (Johnston). What is new is the source: the history sum is the CAL's memory series.